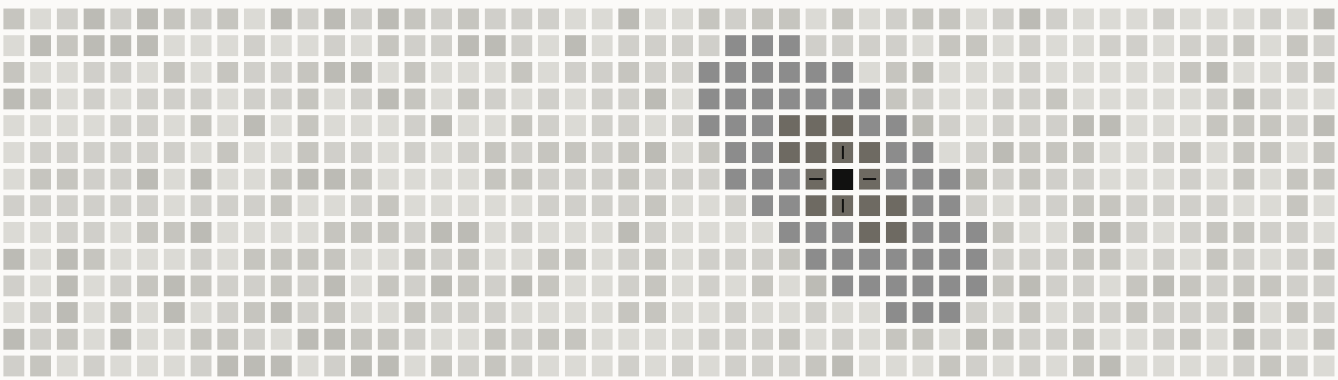
Aurelius

A PREDICTIVE SUPERVISORY LAYER FOR GPU FLEETS

Choose the best fleet trajectory, not just the best next action.

ONE LIVE FLEET STATE · MANY CANDIDATE SEQUENCES · ONE FEASIBLE NEXT ACTION RETURNED

FIG 01 THE SPACE OF ONE DECISION — PER DECISION · READ-ONLY



4,723,920
POSSIBLE COUPLED-POLICY CONFIGURATIONS PER DECISION

72-75 SCORED

1 RETURNED

BOUNDED SEARCH OVER THE REPRESENTABLE SPACE · EACH SQUARE REPRESENTS ONE MULTI-PERIOD COUPLED-POLICY CONFIGURATION — METADATA ONLY

Aurelius adds a pre-execution planning layer above Kubernetes, Slurm, serving schedulers, autoscalers, and fleet-management systems. Before a decision is committed, it generates constraint-aware coupled-policy candidates, advances each against isolated simulated state under the configured constraints, rejects infeasible paths, and returns the highest-scoring feasible policy to the existing control plane. It reads scheduler and fleet metadata only, and never writes directly to the live fleet. In separately tractable spaces, the same bounded-search method matched exhaustive enumeration.

8.24×

mean SLA-safe goodput per modeled infrastructure dollar versus a reconstructed production-class baseline

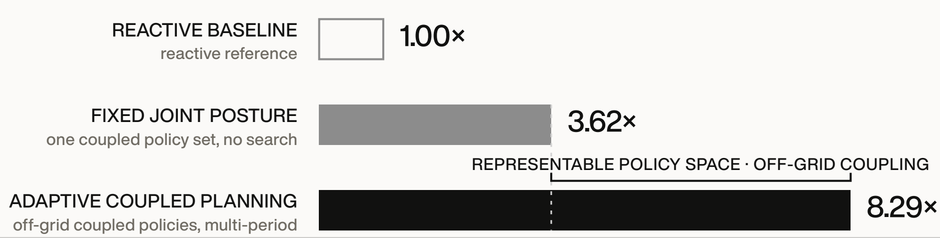
84.5%	fewer GPU-hours under identical input request sets
1,549,249	requests across three selected uncapped high-load windows
3 of 3	windows with lower SLA violation rates

Simulated, load-dependent evidence from selected public-trace windows. Not a production result and not an estimate of performance against a particular fleet or internal scheduler.

The optimizer cannot choose a future it never represents.

The uncapped ablation held the simulator, objective, cost model, constraints, and inputs fixed and varied only the planner's representable policy space and search. It validates the control architecture on four implemented serving controls: batching, capacity, precision, and clock policy. Operator replay would test whether the same planning layer adds value over the broader controls and constraints available in an actual fleet.

FIG 02 THREE STATES OF THE PLANNER — UNCAPPED REPLAY · SINGLE LOAD



THE STEP TO 8.29× IS REPRESENTABLE POLICY SPACE, NOT HARDER SEARCH — RERUN · 07.2026

PROOF · SEARCHING THE FIXED GRID HARDER SATURATES

Exhaustive enumeration of the fixed grid reaches 4.84×; a bounded coupled search matches it at 4.85× while evaluating far fewer candidates. A single control surface (GPU clock only) cannot be scored within the harness budget and lands near 0.69× when forced to complete. The regime-specific candidate set, scored without coupled search, ties the fixed 24-policy grid at 3.62×. Only off-grid coupled policies reach 8.29×, at the lowest SLA violation rate of any arm (0.0022 to 0.0026, versus 0.038 to 0.046 at the reconstructed production-class baseline), across 1,549,249 requests with identical inputs and load in every arm.

The independent rerun reached 8.29× versus the canonical 8.24× benchmark, a 0.6% difference in the mean; individual market results were within 1.2%. **The report retains 8.24× as the canonical headline.**

Uncapped search-strategy ablation, 07.2026: an independent rerun of the exact harness, windows, and load behind the headline benchmark, with production anchors reproducing the published values exactly; ratios are unweighted means across the three markets. The frozen V1 audit separately measured zero search regret where exhaustive enumeration was tractable, at approximately 40 candidate evaluations per decision. Neither validates the simulator or establishes production savings.

Validating the benchmark result on operator data.

FIG 03 UNCAPPED REPLAY · SLA-SAFE GP/$ VS THE RECONSTRUCTED PRODUCTION-CLASS BASELINE THREE MARKETS

1,549,249 total input requests · 84.5% fewer total GPU-hours · lower SLA violation rates in all three windows · both policies received the identical input request sets.

The comparison policy is a reconstructed production-class baseline: a deterministic public-replay policy assembled from established serving and cluster-control mechanisms, including continuous batching, deadline-aware ordering and admission, prefix-cache-aware routing, topology-aware placement, and backlog-driven capacity control. It is stronger than the naive SLA-aware reference, but it is not an operator's internal scheduler and does not reproduce a particular fleet's adaptive logic, internal signals, or cost model.

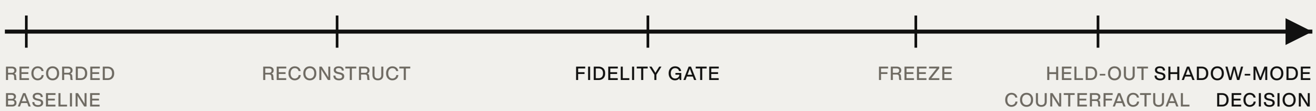
WHAT THIS ESTABLISHES

Under the selected uncapped high-load replay conditions, coupled predictive planning found feasible trajectories with materially better modeled economics and lower SLA violation rates than the reconstructed production-class baseline.

WHAT THIS DOES NOT ESTABLISH

Expected performance against a particular fleet, internal scheduler, workload distribution, or operator cost model. The result is simulated and load-dependent. Its magnitude is materially affected by the benchmark's batching, precision, service-time, and modeled cost assumptions.

Test one bounded historical window.



INPUT	Scheduler and fleet metadata from one representative workload, cluster, or bounded historical window. No customer identity, request payloads, model outputs, proprietary weights, production actuation, or unrestricted infrastructure access.
PROCESS	First reconstruct the recorded control decisions and verify that the simulator reproduces recorded queue, execution, resource-use, and SLA outcomes within agreed tolerances. Counterfactual improvements are scored only if that baseline-reconstruction fidelity gate passes. Then freeze the calibrated simulator and evaluate Aurelius on a held-out historical window using only information available at each decision timestamp.
OUTPUT	Matched SLA, GPU-hour, utilization, completed-work, and agreed economic comparisons; simulator-fidelity results; sensitivity analysis; attribution by active policy dimension; failure cases and negative findings; and a recommendation on whether recommendation-only shadow mode is warranted.

PROPOSED VALIDATION

Use one bounded historical window to estimate how much farther-ahead planning could have improved goodput per dollar under the fleet's actual workload and constraints.

Read-only metadata only. No payloads, production writes, or unrestricted infrastructure access.

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